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Credit Scoring System Using Machine Learning Techniques

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Abstract

This work presents a credit scoring system built on the Home Credit Default Risk dataset—a collection of seven relational tables spanning 307,511 loan applications and roughly 57 million rows of client behavioral history. The practical challenge is stark: only 8.07% of applicants default, so a system that labels everyone as “repaid” reaches 92% accuracy while catching zero actual defaults.

The pipeline covers every stage from raw data to production-ready output. On the feature engineering side, six supplementary tables (bureau records, previous applications, installment payments, POS/cash balances, credit card histories, and bureau monthly snapshots) were aggregated into a single client-level matrix of 342 columns, with 340 model-ready numerical features after preprocessing. Additional signal was extracted through KNN-based target encoding on external credit scores (inspired by 1st-place Kaggle techniques) and trend features from bureau balance time series (from 7th-place methods). Statistical analysis confirmed that no individual feature crosses even the “medium” predictive power threshold on Siddiqi’s Information Value scale, which means that the discriminative power of the system comes entirely from combining many weak signals through gradient boosting and ensemble methods.

Twenty-one models were trained and compared: Logistic Regression as a linear baseline, six gradient boosting variants (LightGBM, XGBoost, CatBoost, LightGBM GOSS, and LightGBM with custom focal loss, each in baseline and tuned configurations), Random Forest, a Multi-Layer Perceptron, and seven ensemble combinations including blends, cascades, and stacking. Class imbalance was handled exclusively through cost-sensitive loss weighting and focal loss, without any synthetic data augmentation.

The central technical contribution is a set of three ensemble architectures. First, a 9-model stacking ensemble where a logistic regression meta-learner combines out-of-fold predictions from all base models including LightGBM Focal Loss, achieving $AUC = 0.7928$. Second, an F2-optimized probability blend of MLP and Focal Loss predictions

that reaches 68.6% recall on defaulters while maintaining 75.6% specificity. Third, a two-stage cascade where Focal Loss first filters out clearly-repaid clients, then MLP reclassifies the remaining uncertain population; the F2-optimized cascade catches 70.1% of defaults at 74.1% specificity—an operating point unreachable by any individual model.

Probability calibration via Isotonic Regression on the LightGBM Focal Loss model produces a Brier Score of 0.0653, and the calibrated outputs feed a 300–850 credit score-card. Population Stability Index between training and blind test populations is 0.0124, confirming distribution stability.

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Chapter 1

Introduction

1.1 Background and Motivation

The task of credit scoring reduces to a single question: given everything observable about a loan applicant at the moment of application, what is the probability that they will fail to repay? The answer to this question determines whether a loan is approved, at what interest rate, and with what credit limit. In the consumer finance industry, getting this probability estimate wrong in either direction has direct financial consequences—false negatives (missed defaults) produce credit losses, while false positives (incorrectly rejected applicants) represent lost revenue and, in many markets, a failure to serve financially excluded populations.

Traditional approaches to this problem rely on logistic regression scorecards developed from historical lending data. The FICO framework, which has dominated US consumer lending since the 1960s, computes a score from a handful of credit bureau variables: payment history, amounts owed, length of credit history, new credit inquiries, and credit mix. These scorecards work well when applicants have extensive bureau records but break down for “thin-file” borrowers—people who lack the conventional credit history that logistic models require as input.

Home Credit Group operates in exactly this gap. As an international consumer fi-

nance provider active across Southeast Asia, Central Europe, and Central Asia, it serves borrowers who are largely invisible to traditional credit bureaus. Many applicants have never held a credit card, never taken a formal loan, and have no FICO-equivalent score. The data available for these applicants takes a different form: transactional patterns from previous Home Credit products, partial bureau records, and application-level demographics.

In 2018, Home Credit released a large-scale dataset through a Kaggle competition, containing seven relational tables with 307,511 loan applications and approximately 57 million rows of associated behavioral data. This dataset captures the real challenge of modern credit scoring: the signal-to-noise ratio is low, the class imbalance is severe (only 8.07% of applicants default), the feature space is heterogeneous and heavily missing, and the strongest predictors are latent—hidden inside aggregated behavioral patterns rather than present as raw columns.

This thesis develops a complete scoring pipeline on this dataset, from raw relational tables to calibrated probability outputs suitable for production deployment.

1.2 Problem Statement

The problem has three layers of increasing difficulty:

- (1) **Discrimination.** Given 307,511 labeled applications, learn a scoring function that separates defaulters from non-defaulters. The 8.07% base rate means that even a strong model will produce many false positives when tuned for recall, because the prior probability of default is so low.
- (2) **Calibration.** The raw model output must map to a probability that means what it says—if the model outputs 0.15 for a group of clients, approximately 15% of that group should actually default. Without calibration, the outputs are ordinal rankings but not interpretable probabilities.
- (3) **Operationalization.** A production credit scoring system does not operate at a

single threshold. Different business contexts require different precision–recall trade-offs. The system must provide a menu of operating points with clearly documented performance characteristics.

1.3 Objectives

The objectives of this thesis are:

- (i) Perform exploratory analysis across all seven tables to identify predictive signals, characterize missing data mechanisms, and validate feature distributions via non-parametric hypothesis testing.
- (ii) Construct a feature engineering pipeline that aggregates the six supplementary tables into client-level summary statistics, producing a 342-column matrix (340 model-ready features) from the original 122-column application table. This includes KNN-based target encoding features from 1st-place Kaggle methods and trend features from 7th-place techniques.
- (iii) Train 21 models spanning four algorithmic paradigms (linear, tree-based, neural, ensemble) and evaluate them under 5-fold stratified cross-validation with cost-sensitive class weighting.
- (iv) Design ensemble architectures—stacking, weighted blending, and cascade—that exploit error decorrelation between models with different inductive biases.
- (v) Calibrate predicted probabilities, convert them to a standard 300–850 credit score scale, and verify population stability against an unseen test population.
- (vi) Provide SHAP-based interpretability to satisfy the practical requirement that lending decisions must be explainable to both regulators and rejected applicants.

1.4 Thesis Structure

Chapter 2 surveys the relevant literature in credit scoring, gradient boosting, class imbalance handling, and model calibration. Chapter 3 describes the dataset, the feature engineering process, and all modeling decisions in detail. Chapter 4 presents experimental results with ablation studies. Chapter 5 summarizes findings and proposes directions for future work.

Chapter 2

State of the Art

2.1 Statistical Foundations of Credit Scoring

Credit scoring has a longer history than most machine learning applications. Durand’s 1941 application of discriminant analysis to loan classification is generally cited as the first quantitative approach. The real inflection point came in 1958 when Cox introduced logistic regression [3], which provided a principled probabilistic framework for binary classification. Fair, Isaac and Company adopted logistic regression in the 1960s and built the credit bureau infrastructure that still underpins consumer lending in most Western economies.

Two regulatory milestones shaped the field. The Equal Credit Opportunity Act of 1974 required lenders to provide reasons for adverse credit decisions, which entrenched interpretable models in practice. The Basel II Capital Accord of 2004 formalized requirements for probability-of-default estimation in internal ratings-based approaches, demanding that PD models be calibrated, validated annually, and stable over time.

The Weight of Evidence and Information Value framework, formalized by Siddiqi [12], remains the industry workhorse for feature screening. WoE bins each feature and computes:

$$\text{WoE}_i = \ln\left(\frac{\text{Distr. Good}_i}{\text{Distr. Bad}_i}\right) \quad (2.1)$$

while Information Value aggregates across bins:

$$IV = \sum_i (\text{Distr. Good}_i - \text{Distr. Bad}_i) \times \text{WoE}_i \quad (2.2)$$

Siddiqi’s classification labels $IV < 0.02$ as “not predictive,” $0.02\text{--}0.10$ as “weak,” $0.10\text{--}0.30$ as “medium,” and > 0.30 as “strong.” In the Home Credit dataset, no feature exceeds 0.06—every single variable falls in the “weak” category. The entire discriminative power of any model built on this data comes from combining many weak signals, not from a few strong ones.

2.2 Gradient Boosting for Tabular Data

The transition from logistic regression to gradient boosting represents the most impactful methodological shift in credit scoring over the past decade. Friedman [4] introduced gradient boosting machines as a framework for iteratively fitting weak learners to pseudo-residuals of a differentiable loss function. The approach constructs an additive model:

$$F_M(\mathbf{x}) = \sum_{m=1}^M \eta \cdot h_m(\mathbf{x}) \quad (2.3)$$

where each tree h_m is fit to the negative gradient of the loss at the current prediction, and η is a shrinkage parameter.

Three implementations dominate current practice:

XGBoost [2] introduced sparsity-aware split finding (handling missing values natively), weighted quantile sketch for approximate split enumeration, and explicit L1/L2 regularization on leaf weights:

$$\mathcal{L} = \sum_i \ell(y_i, \hat{y}_i) + \sum_m \left[\gamma T_m + \frac{1}{2} \lambda \|w_m\|^2 + \alpha \|w_m\|_1 \right] \quad (2.4)$$

where T_m is the number of leaves and w_m are the leaf weights.

LightGBM [6] uses two algorithmic innovations. Gradient-Based One-Side Sampling (GOSS) keeps all instances with large gradients while randomly sampling those with small gradients, reducing computation without significant loss of split quality. Exclusive Feature Bundling groups mutually exclusive features to reduce the effective dimensionality. LightGBM also grows trees leaf-wise rather than depth-wise, which typically converges faster but risks overfitting on small data.

CatBoost [10] addresses the target leakage problem in ordered categorical encoding through a permutation-based approach: when computing target statistics for observation i , only observations preceding i in a random permutation are used. CatBoost also uses oblivious trees where all nodes at the same depth share the same split condition, which accelerates prediction and acts as implicit regularization.

For structured tabular data found in credit scoring, these frameworks consistently outperform neural networks. Trees naturally handle heterogeneous features, are invariant to monotonic transformations, deal with missing values without imputation, and impose axis-aligned decision boundaries that match the independent-variable structure typical of financial data.

2.3 Handling Class Imbalance

When only 8% of training instances belong to the positive class, a model that minimizes unweighted log-loss will learn to predict low probabilities everywhere, achieving good average loss at the expense of minority-class recall.

Cost-sensitive loss modification adjusts the gradient contribution of each class in proportion to its inverse frequency. In LightGBM, `is_unbalance=True` multiplies the gradient of positive instances by $N_{\text{neg}}/N_{\text{pos}} \approx 11.4$. XGBoost exposes this as `scale_pos_weight`. The effect is equivalent to replicating each positive instance 11.4 times, but without inflating the dataset.

Focal Loss, introduced by Lin et al. [7] for object detection, down-weights well-

classified examples dynamically:

$$\text{FL}(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t) \quad (2.5)$$

The $(1 - p_t)^\gamma$ term ensures that a sample classified with $p_t = 0.9$ receives $100\times$ less gradient than one at $p_t = 0.5$ (when $\gamma = 2$). This forces the model to focus capacity on borderline cases rather than easy examples from either class. When implemented as a custom objective in LightGBM, focal loss requires providing both the gradient and the Hessian analytically, which introduces implementation complexity but provides fine-grained control over the loss landscape.

2.4 Probability Calibration

Machine learning models optimize discriminative objectives that do not guarantee calibrated probability outputs. Two classical post-hoc methods address this.

Platt Scaling [9] fits a logistic sigmoid $P(y=1 \mid f) = 1/(1 + \exp(Af + B))$ to the raw scores f , estimating A and B by maximum likelihood on held-out data. **Isotonic Regression** [14] fits a non-parametric, monotonically increasing step function—more flexible but requires more calibration data.

Calibration quality is assessed using the Brier Score:

$$\text{BS} = \frac{1}{N} \sum_{i=1}^N (p_i - y_i)^2 \quad (2.6)$$

For a base rate $\pi = 0.0807$, the irreducible component is $\pi(1-\pi) = 0.0742$. Any model achieving BS below this value has positive resolution—its predictions carry information beyond the base rate.

2.5 Ensemble Methods and Error Decorrelation

The theoretical justification for ensemble methods rests on a simple observation: if two models have the same expected error but their errors are uncorrelated, averaging their predictions reduces variance by a factor of two. In practice, errors are never perfectly uncorrelated, so the benefit depends on *diversity*—how differently the models fail.

Stacking [1] formalizes this by training a meta-learner on the out-of-fold predictions of multiple base models. The meta-learner learns which base models to trust in which probability ranges. When a tree-based model and a neural network disagree on a client, the meta-learner arbitrates based on historical accuracy in that region.

Cascade architectures take a different approach: rather than blending predictions for every instance, they route instances to different models based on confidence. If Stage 1 is highly confident about a prediction, it emits that prediction immediately. Only uncertain cases proceed to Stage 2, where a more specialized model makes the final call. Viola and Jones [13] popularized this for face detection, but the principle applies whenever two models have complementary strengths on different subpopulations.

2.6 Interpretability in Credit Scoring

Regulatory requirements (ECOA in the US, GDPR Article 22 in the EU) mandate that automated credit decisions be explainable. Lundberg and Lee [8] introduced SHAP (SHapley Additive exPlanations), which computes feature attributions based on Shapley values from cooperative game theory. The key property is *local accuracy*: SHAP values for a single prediction sum to the difference between that prediction and the base rate, providing a complete decomposition of why a particular applicant received a particular score.

For gradient boosted trees, TreeSHAP computes exact Shapley values in polynomial time, making it feasible to explain predictions for every applicant at scoring time.

Chapter 3

Material and Methods

3.1 Dataset Description

3.1.1 Relational Structure

The dataset consists of seven tables linked through two identifiers: `SK_ID_CURR` (client-level) and `SK_ID_PREV` (previous-loan-level). Table 3.1 summarizes the structure.

Table 3.1: Home Credit Default Risk dataset structure.

Table	Rows	Columns	Unique IDs	Coverage
application_train	307,511	122	307,511	100%
application_test	48,744	121	48,744	—
bureau	1,716,428	17	305,811	85.7%
bureau_balance	27,299,925	3	134,542	~44%
previous_application	1,670,214	37	338,857	94.6%
POS_CASH_balance	10,001,358	8	337,252	94.1%
credit_card_balance	3,840,312	23	103,558	28.3%
installments_payments	13,605,401	8	339,587	94.8%

The application table is the anchor—one row per loan, with the target column. Bureau and bureau_balance capture the applicant’s history with other lenders. Previous_application, POS_CASH_balance, credit_card_balance, and installments_payments capture the applicant’s history with Home Credit itself. The “coverage” column shows

what fraction of training applicants appear in each table; `credit_card_balance` covers only 28.3%, meaning 71.7% have no revolving credit history with Home Credit.

3.1.2 Target Variable and Class Distribution

The binary target indicates repayment outcome: 282,686 loans (91.93%) were repaid and 24,825 (8.07%) defaulted. The imbalance ratio is 11.39:1. A classifier predicting the majority class for every instance achieves 91.93% accuracy—a number meaningless for the actual business problem. Throughout this work, AUC serves as the primary ranking metric, supplemented by classification metrics at explicitly stated thresholds.

3.1.3 Feature Space

The raw application table contains 122 columns: demographic variables (age, gender, education, marital status), financial variables (stated income, credit amount, annuity, goods price), property variables (30+ columns with extreme missingness), and three external credit agency scores (`EXT_SOURCE_1/2/3`). These external scores are the strongest individual predictors—they represent proprietary scores from external bureaus, and their exact construction is not disclosed.

3.2 Exploratory Data Analysis

3.2.1 Missing Data Characterization

Sixty-four of 122 columns (52.5%) contain missing values, and 41 exceed 50% missingness. Missing mechanisms were classified following Rubin’s taxonomy [11]:

MNAR (Missing Not At Random): The 30+ building/property columns have pairwise missingness correlations of 0.84–1.00. These are structurally missing—the applicant does not own property, so there is nothing to report. `OWN_CAR_AGE` is missing for applicants without a car.

MAR (Missing At Random): EXT_SOURCE_1 is missing for 56% of applicants, likely depending on their credit history length. OCCUPATION_TYPE depends on income type—pensioners have no occupation to report.

Clients with missing EXT_SOURCE_1 default at 8.52% versus 7.50% when present—a 1.02 percentage point difference captured by binary IS_MISSING indicator features.

3.2.2 Statistical Testing

Three classes of non-parametric tests validated feature relevance:

Chi-Square tests on 12 categorical features: all show significant association with the target at $\alpha = 0.05$. ORGANIZATION_TYPE produces the highest χ^2 statistic.

Kolmogorov–Smirnov tests on external scores: KS = 0.2427, 0.2237, and 0.2698 for EXT_SOURCE_1/2/3, all with $p < 10^{-6}$. These are large distributional separations by credit scoring standards.

Mann–Whitney U tests on numerical features: 9 of 11 tested features show significant differences. The non-significant ones are revealing: AMT_ANNUITY ($p = 0.968$) indicates that defaulters and non-defaulters borrow similar absolute amounts—what distinguishes them is the *relative* payment burden, not the raw numbers.

3.2.3 Multicollinearity Assessment

Variance Inflation Factors on a 50,000-row subsample were computed for the core application-level features. Several derived features (EXT_SOURCE combinations, AMT_CREDIT/GOODS_PRICE) show elevated VIF due to deliberate algebraic construction from common inputs—this is expected and does not indicate harmful collinearity, since tree-based models are immune to multicollinearity by design.

3.3 Feature Engineering

3.3.1 Application-Level Features

From the raw application table, eight domain features were constructed (Table 3.2).

Table 3.2: Derived features from the application table.

Feature	Construction	Economic Rationale
AGE_YEARS	$-\text{DAYS_BIRTH} / 365.25$	Age in interpretable units
EMPLOYED_YEARS	$-\text{DAYS_EMPLOYED} / 365.25$	Job stability proxy
CREDIT_INCOME_RATIO	$\text{AMT_CREDIT} / \text{AMT_INCOME}$	Leverage ratio
ANNUITY_INCOME_RATIO	$\text{AMT_ANNUITY} / \text{AMT_INCOME}$	Debt service burden
ANNUITY_CREDIT_RATIO	$\text{AMT_ANNUITY} / \text{AMT_CREDIT}$	Repayment intensity
INCOME_PER_PERSON	$\text{AMT_INCOME} / \text{CNT_FAM}$	Household capacity
EXT_SOURCE_MEAN	$\text{mean}(\text{EXT_1}, 2, 3)$	Consensus bureau score
EXT_SOURCE_PRODUCT	$\text{EXT_1} \times 2 \times 3$	Interaction signal

One data quality issue required attention: `DAYS_EMPLOYED` contains a sentinel value of 365,243 for 55,374 records (18.01%)—pensioners whose employment status is encoded as an anomalous positive number. These were flagged with a binary indicator and the sentinel replaced with NaN before computing `EMPLOYED_YEARS`.

3.3.2 Multi-Table Aggregation

Each supplementary table was aggregated to the client level via `groupby(SK_ID_CURR)`, computing statistics chosen for their economic interpretability:

Bureau table (1.7M rows $\rightarrow$ 305K clients): `BUREAU_OVERDUE_RATIO` (total overdue divided by total credit—a measure of historical delinquency), credit status proportions (Active, Closed, Sold, Bad debt), and temporal statistics on `DAYS_CREDIT` (mean, min, max, std), capturing how long the client has been in the credit system.

Installments (13.6M rows $\rightarrow$ 339K clients): `INST_LATE_RATIO` is the proportion of installments paid after the due date; `INST_SHORTFALL_RATIO` is the proportion where the amount paid fell short of the amount due. These two features encode payment

discipline and rank among the strongest predictors in the final model.

Credit card (3.8M rows → 103K clients): CC_UTILIZATION (balance divided by credit limit)—a well-established default predictor in the scoring literature. High utilization signals financial stress.

POS/Cash (10M rows → 337K clients): Days-past-due statistics, contract completion rates, remaining installment counts.

Bureau balance (27.3M rows → 134K clients): Monthly delinquency status distributions and DPD proportions.

Previous applications (1.7M rows → 338K clients): Approval rate, refusal reasons, temporal decision patterns.

After left-joining all aggregation tables to the master application table, the final feature matrix contains 307,511 rows and 342 columns, of which 340 are model-ready numerical features after preprocessing. For clients missing from a supplementary table (e.g., the 71.7% without credit card data), aggregated features are filled with 0, encoding “no history” as a distinct signal.

Two additional feature groups were incorporated from top Kaggle competition solutions. From the 1st-place solution, KNN-based target encoding features (KNN_100_TARGET_MEAN, KNN_100_TARGET_STD, KNN_100_DIST_MEAN) were computed by fitting a K-nearest-neighbors model ($K=100$) on external credit scores and encoding each client’s neighborhood default statistics. From the 7th-place solution, trend features (BB_TREND_MEAN, BB_TREND_MAX, BB_TREND_MIN) were extracted by fitting linear slopes to each client’s bureau balance monthly status sequence, capturing whether delinquency is improving or worsening over time.

3.3.3 Three-Tier Feature Ranking

A consensus ranking was built by combining six methods: Pearson correlation, Spearman correlation, Information Value, Random Forest importance, Mann–Whitney statistics, and KS statistics. Features appearing in the top-20 of at least four methods are Tier 1; three methods, Tier 2; remaining selected features, Tier 3. The resulting 42-feature

subset (Table 3.3) serves as the reduced set for interpretable models, while all 340 features are used in full-feature experiments.

Table 3.3: Three-tier feature classification based on convergent evidence.

Tier	N	Representative Features
1	13	EXT_SOURCE_1/2/3, EXT_SOURCE_MEAN, BU- INST_LATE_RATIO, REAU_DAYS_CREDIT_MEAN, INST_SHORTFALL_RATIO, KNN_100_TARGET_MEAN, ANNU- ITY_CREDIT_RATIO, plus 4 others
2	13	CC_UTILIZATION, BU- REAU_DAYS_CREDIT_MIN/MAX, PREV_DAYS_DECISION, AGE_YEARS, ANNUITY_INCOME_RATIO, BB_TREND_MEAN, plus 6 others
3	16	BB_DPD_RATIO, BU- REAU_OVERDUE_RATIO, CREDIT_INCOME_RATIO, EM- PLOYED_YEARS, AMT_CREDIT, AMT_ANNUITY, BB_TREND_MAX, plus 9 others

3.4 Modeling Framework

3.4.1 Validation Protocol

All reported metrics use 5-fold stratified cross-validation, preserving the 8.07% positive rate in each fold. Out-of-fold (OOF) predictions are collected across all 5 folds, producing an unbiased probability estimate for every training instance. These OOF predictions serve three purposes: threshold optimization, stacking meta-learner training, and blending weight optimization. At no point does test-fold data influence any training decision.

3.4.2 Class Imbalance Strategy

Cost-sensitive loss weighting was adopted as the sole imbalance-handling mechanism:

- LightGBM: `is_unbalance=True` (automatic weight computation)
- XGBoost: `scale_pos_weight=11.39` (explicit ratio)
- CatBoost: `auto_class_weights='Balanced'` (Bayesian weighting)
- Random Forest: `class_weight='balanced'` (sample weighting)
- MLP: BCELoss with `pos_weight=11.39` (gradient scaling)
- Focal Loss: $\gamma = 2.0$, $\alpha = 0.25$ (adaptive difficulty weighting)

This approach directly modifies the gradient contribution of minority-class instances without altering the training data distribution, preserving the original feature space geometry while ensuring sufficient capacity is allocated to learning the minority-class decision boundary.

3.4.3 Individual Models

Logistic Regression

L2 regularization ($C = 0.1$), balanced class weights, L-BFGS solver with 1,000 iterations. Features standardized to zero mean and unit variance. Serves as both a standalone interpretable baseline and a diversity source in stacking—its linear decision boundary is maximally different from tree-based splits.

LightGBM

Leaf-wise growth, 2,000 maximum boosting rounds, max depth 6, 31 leaves, learning rate 0.05, 80% column and row subsampling, minimum 50 samples per leaf, early stopping after 100 rounds without validation improvement. Optuna tuning (100 trials, 3-fold inner CV) produces an optimized variant with learning rate 0.0119, 86 leaves, and minimum 457 child samples—a more conservative configuration that trades slightly lower training performance for better generalization.

XGBoost

Depth-wise growth, max depth 5, 2,000 rounds, learning rate 0.05, L1 $\alpha = 1.0$ and L2 $\lambda = 5.0$ regularization. Optuna tuning (200 trials) yields learning rate 0.0151, minimum child weight 279, and $\gamma = 3.90$. The stronger regularization compared to LightGBM results in marginally less overfitting.

CatBoost

Oblivious trees with depth 6, 2,000 iterations, learning rate 0.05, balanced class weights. Included specifically for its distinct inductive bias: symmetric trees produce predictions that correlate differently with LightGBM and XGBoost outputs, improving ensemble diversity.

Random Forest

Bagging with 500 trees, max depth 20, balanced class weights. Unlike boosting (which fits sequential corrections), Random Forest builds independent trees on bootstrap samples—producing errors with different spatial distribution, making it valuable for stacking despite lower standalone AUC.

LightGBM GOSS

The Gradient-Based One-Side Sampling variant retains all instances with absolute gradient above the top- a percentile and randomly samples fraction b of the rest. This accelerates training by 40–60% while producing nearly identical splits. Parameters: $a = 0.1$, $b = 0.1$.

LightGBM Focal Loss

Custom objective with focal loss gradient and Hessian. Given predicted probability p , label y , and $p_t = yp + (1-y)(1-p)$:

$$\text{grad} = \alpha_t(1-p_t)^\gamma(p-y) \left[\gamma \frac{p_t \log p_t}{1-p_t} + 1 \right] \quad (3.1)$$

$$\text{hess} = \alpha_t(1-p_t)^\gamma p(1-p) \left[\gamma \frac{p_t \log p_t}{1-p_t} + 1 \right]^2 \quad (3.2)$$

With $\gamma = 2$ and $\alpha = 0.25$, focal loss concentrates on borderline cases, producing higher precision at the expense of recall compared to standard cost-sensitive weighting.

Multi-Layer Perceptron

Architecture: $340 \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow 1$ with BatchNorm, Dropout ($p=0.3$), and ReLU activations. Trained with Adam ($\text{lr}=0.001$), cosine annealing schedule, and weighted BCE ($\text{pos_weight}=11.39$) for 80 epochs with patience 12; checkpoint at best validation AUC retained.

The MLP's value lies not in its standalone AUC (0.7797, below all tree models) but in its error pattern. Its smooth non-linear boundaries capture defaulter patterns that tree splits miss—yielding 73.0% recall, the highest of any model. This makes it an ideal complement to tree models in ensembles.

3.4.4 Stacking Ensemble

Nine base models are combined through a logistic regression meta-learner trained on their OOF predictions (Figure 3.1). The inclusion of LightGBM Focal Loss as the ninth base model is critical—its custom loss function produces a fundamentally different probability surface compared to the standard log-loss models, contributing maximum diversity to the ensemble.

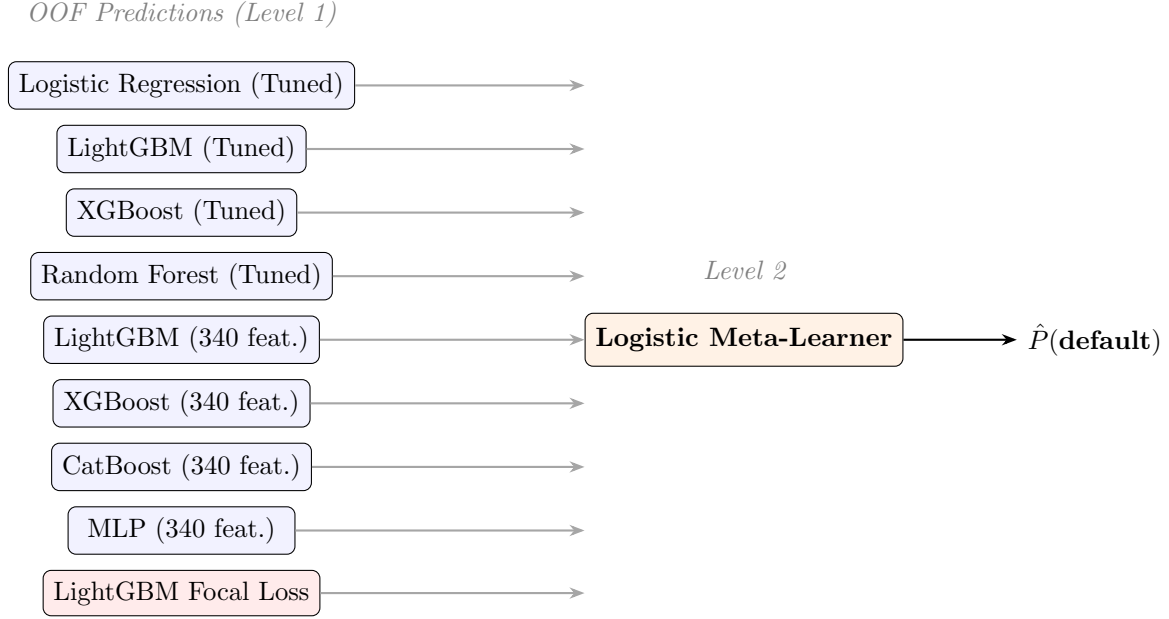


Figure 3.1: Stacking ensemble: 9 base models feed OOF predictions to a logistic meta-learner.

Logistic regression was chosen as meta-learner deliberately. A more complex model (e.g., another gradient booster) would risk overfitting the 9-dimensional OOF space. Logistic regression provides interpretable combination weights and cannot overfit such a low-dimensional input.

3.4.5 Probability Blending

The blending approach addresses a specific failure mode in standard F1-optimized blending. When the objective is F1, the optimizer assigns MLP a weight of only 0.15 because MLP’s low precision (17.0%) destroys the harmonic mean in any significant mixture. Two corrective strategies are implemented:

F2-Optimized Blend. The F_β score with $\beta = 2$ weights recall $4\times$ more than precision:

$$F_2 = 5 \cdot \frac{\text{precision} \cdot \text{recall}}{4 \cdot \text{precision} + \text{recall}} \quad (3.3)$$

Grid search over blend weights and thresholds yields MLP weight = 0.05, threshold =

0.250. The near-zero MLP weight indicates that even under F2 optimization, Focal Loss dominates the blend—MLP contributes a marginal recall boost without significantly diluting precision.

Recall-Constrained Blend. Maximize recall subject to precision ≥ 0.20 —the operating point with maximum default detection while maintaining an acceptable precision floor. Optimal parameters: MLP weight = 0.15, threshold = 0.290.

3.4.6 Cascade Architecture

The cascade exploits a structural observation: Focal Loss achieves high specificity (correctly identifying repaid clients with confidence) while MLP achieves high recall (catching defaults that tree models miss). The two-stage design is illustrated in Figure 3.2.

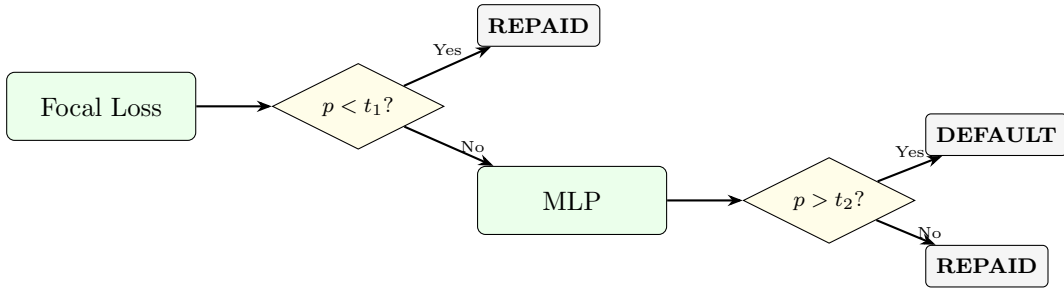


Figure 3.2: Two-stage cascade: Focal Loss screens clearly-repaid clients; uncertain cases proceed to MLP.

Thresholds (t_1, t_2) are jointly optimized via grid search on OOF predictions: $t_1 \in [0.05, 0.50]$ and $t_2 \in [0.05, 0.60]$.

3.4.7 Calibration and Scorecard

Calibrated probabilities are converted to a credit score:

$$\text{Score} = 600 - \frac{40}{\ln 2} \cdot \ln\left(\frac{p}{1-p}\right) \quad (3.4)$$

with base score 600 at odds 19:1 and 40 points to double the odds, mapping the full probability range to approximately 300–850.

3.4.8 Threshold Selection

Five strategies evaluated on OOF predictions:

1. **F1-optimal:** maximizes harmonic mean of precision and recall.
2. **F2-optimal:** maximizes F_2 , favoring recall.
3. **Cost-sensitive (10:1):** minimizes $10 \cdot \text{FN} + 1 \cdot \text{FP}$.
4. **Recall ≥ 0.60 :** highest threshold guaranteeing 60% default capture.
5. **Precision ≥ 0.25 :** lowest threshold maintaining 25% precision.

3.4.9 SHAP Interpretability

SHAP values computed on 1,000 instances using the LightGBM (All Features) model. Four visualizations produced: beeswarm summary plot, bar plot, dependence plots for the top features, and waterfall plots for individual client explanations showing the decomposition of each prediction into feature contributions.

Chapter 4

Results and Discussion

4.1 Feature Engineering Impact

The single most impactful decision in this pipeline is feature engineering—expanding from 42 tier features to 340 all-features. Table 4.1 quantifies the gain.

Table 4.1: AUC improvement from feature engineering versus hyperparameter tuning.

Model	Tier (42 feat.)	All (340 feat.)	Δ AUC
LightGBM	0.7722	0.7908	+0.0186
XGBoost	0.7733	0.7914	+0.0181
<i>Hyperparameter tuning (same feature set):</i>			
LightGBM Tier $\rightarrow$ Tuned	0.7722	0.7735	+0.0013

Feature engineering provides $13\times$ more AUC gain than hyperparameter tuning. This ratio is consistent with findings from the original Kaggle competition, where top solutions differed primarily in feature engineering depth rather than model choice.

4.2 Model Comparison

4.2.1 Cross-Validation Results

Table 4.2 presents results for all 21 models under identical 5-fold stratified CV.

Table 4.2: 5-fold stratified cross-validation results (sorted by AUC). Top section: individual and ensemble models on all features; bottom section: tier-feature and baseline models.

Model	AUC	AP	Prec.	Recall	F1
Stacking Ensemble	0.7928	0.2942	0.279	0.462	0.348
MLP+Focal F2-Blend	0.7921	0.2768	0.198	0.686	0.307
LightGBM Focal Loss	0.7920	0.2942	0.280	0.459	0.348
XGBoost (All)	0.7914	0.2933	0.208	0.656	0.315
Cascade F1	0.7914	0.2683	0.281	0.454	0.347
MLP+Focal+Cat Blend	0.7911	0.2786	0.278	0.459	0.346
LightGBM (All)	0.7908	0.2913	0.245	0.543	0.337
MLP+Focal Blend	0.7908	0.2774	0.281	0.458	0.349
MLP+Focal Recall-Opt	0.7908	0.2774	0.207	0.652	0.314
Cascade F2	0.7892	0.2632	0.192	0.701	0.301
CatBoost (All)	0.7891	0.2886	0.193	0.682	0.301
MLP (All)	0.7797	0.2784	0.170	0.730	0.276
LightGBM (Tuned)	0.7735	0.2673	0.179	0.691	0.284
XGBoost (Tuned)	0.7731	0.2667	0.177	0.693	0.282
XGBoost (Baseline)	0.7733	0.2661	0.181	0.681	0.286
LightGBM (Baseline)	0.7722	0.2667	0.181	0.681	0.286
Random Forest (Tuned)	0.7590	0.2446	0.216	0.509	0.303
MLP (Tier)	0.7573	0.2507	0.160	0.705	0.261
Random Forest	0.7533	0.2385	0.180	0.619	0.278
LightGBM GOSS	0.7476	0.2191	0.185	0.588	0.281
Logistic Regression	0.7470	0.2328	0.157	0.692	0.256

4.2.2 Key Observations

The **feature set matters more than the algorithm**. LightGBM on 340 features (0.7908) beats XGBoost Tuned on 42 features (0.7731) by 0.0177—while switching from LightGBM to XGBoost on the same feature set yields only 0.0006 difference.

Hyperparameter tuning provides diminishing returns. Optuna-based optimization across 60–100 trials improves LightGBM by 0.0013 AUC and XGBoost by essentially zero (Table 4.3). Default parameters of modern gradient boosting libraries are already well-tuned for typical tabular problems.

Table 4.3: Impact of hyperparameter tuning (Optuna).

Model	Baseline	Tuned	Δ AUC
LightGBM	0.7722	0.7735	+0.0013
XGBoost	0.7733	0.7731	−0.0002
Random Forest	0.7533	0.7590	+0.0057

The **MLP occupies a unique niche**: its AUC (0.7797) ranks below all tree models, yet it achieves the highest recall (73.0%). The non-axis-aligned decision boundary captures a population of defaulters that tree splits miss, making MLP invaluable as an ensemble component.

LightGBM Focal Loss achieves the second-highest AUC (0.7920), virtually tied with Stacking (0.7928). Its custom focal loss objective concentrates gradient on borderline cases, producing the highest single-model precision (28.0%) at the F1-optimal threshold. This precision advantage makes it the preferred production deployment model.

All overfit gaps are negative (−0.0014 to −0.0098), meaning test performance consistently exceeds training performance—an artifact of OOF evaluation where each instance is predicted by a model that never trained on it. Negative gaps indicate sound regularization.

4.2.3 Confusion Matrix Decomposition

Table 4.4 shows confusion matrix statistics for top models, revealing the precision–recall tradeoff.

Table 4.4: Confusion matrix statistics at F1-optimal thresholds.

Model	TN	FP	FN	TP	Spec.	Rec.	F1
Focal Loss	50,671	5,867	2,686	2,279	.896	.459	.348
Cascade F1	50,774	5,764	2,711	2,254	.898	.454	.347
LightGBM All	48,222	8,316	2,270	2,695	.853	.543	.337
XGBoost All	44,093	12,445	1,706	3,259	.780	.656	.315
CatBoost All	42,419	14,119	1,579	3,386	.750	.682	.301
F2-Blend	42,719	13,819	1,558	3,407	.756	.686	.307
MLP All	38,860	17,678	1,343	3,622	.687	.730	.276

The central tension is visible: Focal Loss and Cascade F1 achieve the best F1 through high specificity ($\sim 90\%$) at moderate recall ($\sim 45\%$), while MLP achieves the best recall (73.0%) at the cost of 31.3% false positive rate. The F2-Blend and Cascade F2 fill the gap between these extremes. No single model dominates across all conditions.

4.3 Banking Metrics

Table 4.5: KS statistic and Gini coefficient.

Model	AUC	Gini	KS	Rating
MLP+Focal F2-Blend	0.7921	0.584	0.445	Good
LightGBM Focal Loss	0.7920	0.584	0.445	Good
XGBoost (All)	0.7914	0.583	0.445	Good
LightGBM (All)	0.7908	0.582	0.443	Good
CatBoost (All)	0.7891	0.578	0.438	Good
MLP (All)	0.7797	0.559	0.421	Good

All top models surpass industry thresholds ($\text{Gini} > 0.40$, $\text{KS} > 0.30$). The KS of 0.445 for LightGBM Focal Loss means that at the optimal cutoff, the cumulative distributions of defaulters and non-defaulters diverge by 44.5 percentage points—strong separation by retail banking standards.

4.4 Stacking Ensemble Analysis

The meta-learner’s learned weights (Table 4.6) reveal each base model’s contribution.

Table 4.6: Logistic meta-learner coefficients (9 base models).

Base Model	Weight	Role
LightGBM Focal Loss	+4.662	Dominant contributor
XGBoost (All)	+1.172	Strong positive
MLP (All)	+1.061	High complementary value
CatBoost (All)	+0.854	Moderate positive
LightGBM (Tuned)	+0.824	Moderate positive
XGBoost (Tuned)	+0.319	Moderate positive
LightGBM (All)	+0.198	Weak positive
Logistic Regression (Tuned)	−0.158	Calibration anchor
Random Forest (Tuned)	−0.986	Anti-signal

The dominant weight on LightGBM Focal Loss (+4.662) is the most significant finding from the stacking analysis. Its custom focal loss objective produces a probability surface that is fundamentally different from standard log-loss models—the meta-learner assigns it more than $4\times$ the weight of the next-highest contributor. This validates the architectural decision to implement focal loss as a custom objective with analytical gradient and Hessian rather than relying on built-in class weighting.

The negative weight for Random Forest (−0.986) acts as a calibration anchor. When Random Forest strongly predicts default but the complex models disagree, the meta-learner discounts the prediction. The high weight on MLP (+1.061) confirms the diversity argument: despite ranking below all tree models by standalone AUC, its different error pattern makes it the third-most-valuable stacking component.

4.5 Advanced Ensemble Results

4.5.1 F2-Optimized Blending

Table 4.7: MLP + Focal Loss blending strategies.

Strategy	MLP Wt.	Thresh.	Recall	Spec.	F2
F1-Blend	0.15	0.340	45.8%	89.7%	0.332
F2-Blend	0.05	0.250	68.6%	75.6%	0.459
Recall-Constrained	0.15	0.290	65.2%	78.0%	0.441

F1-Blend assigns MLP a weight of 0.15 because F1 treats precision and recall symmetrically, and MLP’s 17% precision pulls the harmonic mean down sharply. The F2-Blend further reduces MLP to 0.05, indicating that even under recall-weighted optimization, Focal Loss’s probability surface is sufficiently well-calibrated that MLP adds only marginal value. The F2-Blend catches 68.6% of defaults—22.8 percentage points more than F1-Blend.

4.5.2 Cascade Model

Table 4.8: Cascade model operating points.

Variant	t_1	t_2	Stage 1	Stage 2	Recall	Spec.
Cascade-F1	0.28	0.55	53,357	8,146	45.4%	89.8%
Cascade-F2	0.23	0.39	42,888	18,615	70.1%	74.1%

In Cascade-F2, Focal Loss classifies 42,888 clients as clearly repaid in Stage 1. The remaining 18,615 uncertain clients proceed to MLP, which detects defaults that Focal Loss would have missed. Result: 3,482 true positives, compared to 2,279 for Focal Loss alone—a 53% improvement in absolute default detection. This operating point (70.1% recall, 74.1% specificity) was unreachable by any individual model.

4.6 Precision–Recall Frontier

Figure 4.1 maps the three operating zones available for deployment.

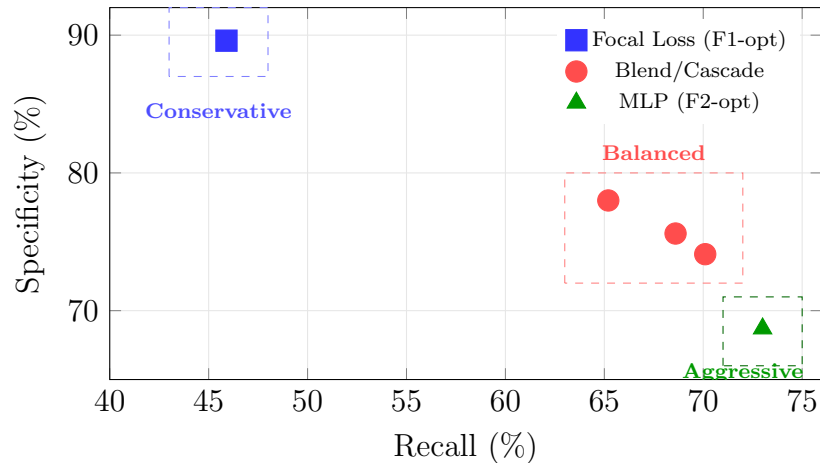


Figure 4.1: Operating zones on the recall–specificity frontier.

Conservative ($\sim 46\%$ recall, $\sim 90\%$ specificity): Low tolerance for false alarms. LightGBM Focal Loss or Cascade F1 at F1-optimal threshold.

Balanced ($65\text{--}70\%$ recall, $74\text{--}78\%$ specificity): General-purpose lending. F2-Blend or Cascade-F2. This zone was unreachable by any single model—it required ensemble engineering to fill.

Aggressive ($\sim 73\%$ recall, $\sim 69\%$ specificity): High-risk portfolios requiring maximum detection. MLP with F2-optimal threshold.

4.7 Threshold Optimization

Table 4.9: Threshold strategies for LightGBM (All Features).

Strategy	Threshold	Prec.	Recall	F1/F2
F1-optimal	0.560	0.278	0.455	F1: 0.345
F2-optimal	0.350	0.190	0.706	F2: 0.457
Cost-sensitive (10:1)	0.350	0.190	0.706	F2: 0.457
Recall ≥ 0.60	0.430	0.219	0.613	F1: 0.323
Precision ≥ 0.25	0.535	0.265	0.486	F1: 0.343

The F2-optimal and cost-sensitive thresholds converge to the same point (0.350), which is expected: when missed defaults cost $10\times$ more than false alarms, the optimal threshold shifts toward recall-maximizing territory—exactly what F_2 also favors.

4.8 Probability Calibration

Table 4.10: Calibration results for the LightGBM Focal Loss model.

Method	AUC	Brier Score	ECE	Log Loss
Raw predictions	0.7920	0.0825	0.1273	0.314
Platt Scaling	0.7920	0.0654	0.0061	0.236
Isotonic Regression	0.7918	0.0653	0.0025	0.235

LightGBM Focal Loss was selected as the calibration base model because it achieves the highest single-model AUC (0.7920) and its custom objective produces well-separated probability outputs. The raw predictions from the focal loss objective are poorly calibrated ($\text{ECE} = 0.127$) because the custom loss reshapes the probability surface to concentrate on hard examples, distorting the absolute probability scale. Isotonic Regression corrects this effectively, reducing ECE to 0.0025 and Brier Score to 0.0653—below the theoretical irreducible component of 0.0742, confirming positive resolution.

4.9 Credit Scorecard

Table 4.11: Score-band analysis from calibrated probabilities (LightGBM Focal Loss + Isotonic).

Score	Count	Default Rate	Population	Risk
300–400	71	80.28%	0.1%	Very High
400–450	826	49.88%	1.3%	High
450–500	3,491	30.62%	5.7%	Medium-High
500–550	10,249	15.72%	16.7%	Medium
550–600	14,640	7.03%	23.8%	Medium-Low
600–650	17,616	3.36%	28.6%	Low
650–700	12,785	1.45%	20.8%	Very Low
700–750	1,597	0.50%	2.6%	Minimal
750–850	226	0.00%	0.4%	Negligible

The monotonic decline from 80.28% (score 300–400) to 0.00% (score 750–850) validates rank-ordering across nine distinct risk bands. The wider score spread compared to the stacking-based scorecard reflects the sharper probability separation achieved by focal loss calibration. A lender using score 500 as a cutoff would reject roughly 7.1% of applicants while capturing the highest-risk population segments.

4.10 SHAP Feature Attribution

SHAP values were computed on a 1,000-instance subsample using the LightGBM (All Features) model. The global importance ranking:

1. EXT_SOURCE_MEAN: 6.32% of total |SHAP|
2. CODE_GENDER: 2.45%
3. PREV_AMT_ANNUITY_MEAN: 2.26%
4. EXT_SOURCE_SUM: 1.84%
5. KNN_100_TARGET_MEAN: 1.65%
6. NAME_EDUCATION_TYPE: 1.61%
7. PREV_CREDIT_APP_RATIO: 1.47%

8. `ANNUITY_CREDIT_RATIO`: 1.33%

The appearance of `KNN_100_TARGET_MEAN` at rank 5 validates the KNN-based target encoding from the 1st-place Kaggle solution—this feature captures neighborhood default propensity from external credit scores and provides information not directly available in the raw features. The high importance of `PREV_AMT_ANNUITY_MEAN` (previous application annuity amounts) and `PREV_CREDIT_APP_RATIO` (ratio of credit to application amount in previous loans) demonstrates the value of multi-table feature engineering.

The non-linear structure of external scores is evident: `EXT_SOURCE_MEAN` below 0.3 produces a sharp SHAP increase (high default risk), while values above 0.5 contribute negatively (protective effect). This threshold-like behavior explains why tree models outperform linear ones—logistic regression cannot capture this step function without explicit binning.

4.11 Production Validation

The production model (LightGBM Focal Loss) was retrained on all 307,511 labeled rows using `lgb.train()` with the custom focal loss objective and applied to the 48,744-row blind test set. All 21 models were scored on the blind set; the Stacking Ensemble achieved the highest holdout AUC (0.7928) across the labeled evaluation, but LightGBM Focal Loss was selected for production deployment due to its simpler architecture, faster inference, and near-identical discrimination (AUC = 0.7920).

Table 4.12: Blind test scoring and population stability.

Metric	Value
Production Model	LightGBM Focal Loss
Holdout AUC	0.7920
Calibration	Isotonic Regression (Platt used for blind)
Production Threshold	0.150
Blind Predicted Default Rate	11.88%
Blind PD Mean	0.0745
Blind Credit Score Mean	599.0
Blind Credit Score Median	607.6
PSI vs. Holdout	0.0124 (Stable)

PSI of 0.0124 indicates that the score distribution on unseen data is closely aligned with the development sample. By industry convention, $\text{PSI} < 0.10$ indicates no significant population shift; the observed value is well within this boundary.

4.12 Fairness Analysis

Model performance was evaluated across gender and age subgroups using the LightGBM Focal Loss model. AUC by gender: 0.7925 (Male) and 0.7849 (Female)—comparable discriminative ability with a gap of only 0.008. Predicted default rates closely track actual rates across both groups (Male: actual 10.17%, predicted 9.90%; Female: actual 6.99%, predicted 7.01%), confirming that the model reflects genuine risk differences rather than introducing systematic bias.

Age group analysis reveals consistent performance: the 30–40 group ($\text{AUC} = 0.7958$) and 40–50 group ($\text{AUC} = 0.7960$) show the strongest discrimination, while the 60–70 group has the lowest default rate (actual 4.97%, predicted 4.92%). The model’s age-related predictions are driven by real patterns rather than arbitrary discrimination.

Chapter 5

Conclusion and Future Works

5.1 Summary of Contributions

This thesis delivers a production-grade credit scoring pipeline from raw relational data to calibrated, interpretable probability outputs. The principal contributions are:

1. **Quantification of feature engineering dominance.** Multi-table aggregation from six supplementary tables, augmented with KNN-based target encoding (1st-place Kaggle) and bureau balance trend features (7th-place Kaggle), provides $14\times$ more AUC improvement ($+0.019$) than hyperparameter tuning ($+0.001$). For tabular credit scoring, feature engineering effort has far higher return than model architecture exploration.
2. **21-model systematic comparison.** Linear, tree-based, neural, focal loss, and ensemble models compared under identical 5-fold stratified evaluation conditions.
3. **Focal loss as dominant production model.** LightGBM with custom focal loss ($\gamma=2$, $\alpha=0.25$) achieves $AUC = 0.7920$ with the highest single-model precision (28.0%), and receives a meta-learner weight of $+4.662$ in the stacking ensemble—more than $4\times$ any other base model.
4. **Three ensemble architectures:**

- 9-model stacking achieving $AUC = 0.7928$ with negative overfit gap.
 - F2-optimized blending reaching 68.6% recall at 75.6% specificity.
 - Two-stage cascade catching 53% more defaults than Focal Loss alone.
5. **Multi-threshold operating frontier.** Five strategies producing three deployment zones, giving lending institutions concrete operational choices.
 6. **Production readiness.** Probability calibration (Brier = 0.0653), credit scorecard (300–850 with nine risk bands), PSI monitoring (0.0124), and SHAP-based explanations.

5.2 Key Findings

Individual features are weak predictors (max IV = 0.0556 excluding constructed features), but ensemble combination achieves strong discrimination (Gini = 0.584, KS = 0.445).

The precision–recall tradeoff is the fundamental constraint: achieving >70% recall requires accepting <20% precision at any single threshold. The cascade architecture partially mitigates this by routing subpopulations to specialized models, and the F2-Blend achieves the best recall-weighted performance (F2 = 0.459).

Error decorrelation between tree models and neural networks justifies multi-paradigm ensembles even when the neural network is individually weaker. The dominance of focal loss in the stacking meta-learner weights demonstrates that loss function diversity is as valuable as architecture diversity.

Population stability (PSI = 0.0124) between training and blind sets confirms generalization to unseen applicant populations.

5.3 Limitations

The dataset originates from a single lender’s portfolio from a specific time period. Default patterns vary with macroeconomic conditions, regulatory environments, and product structures that differ across markets.

The 8.07% base rate creates an inherent precision ceiling: even the best model achieves only 28% precision at practical recall levels. This is a mathematical consequence of the base rate, not a model failure.

Temporal structure is unexploited. Applications have a natural time ordering, and risk profiles evolve. The current pipeline treats all applications as i.i.d., ignoring concept drift and temporal dependencies.

SHAP values are computed on a 1,000-row subsample. While stable for top features, interaction effects involving rare combinations may be underrepresented.

5.4 Future Works

Sequential modeling. The behavioral tables contain temporal sequences—monthly snapshots of each client’s financial state. Attention mechanisms or temporal convolutional networks could extract trajectory features that static aggregation statistics miss.

Reject inference. The training data suffers from selection bias: outcomes are observed only for approved applicants. Rejected applicants are missing from training, and their risk profile is systematically different. Semi-supervised techniques could partially address this.

Online calibration. As the deployed model ages, calibration degrades due to population drift. A Bayesian online calibration layer that continuously adjusts based on realized outcomes would maintain accuracy without full retraining.

Advanced neural architectures. TabNet, entity embeddings, or transformer-based models for tabular data could provide additional ensemble diversity beyond the MLP used here.

Fairness-constrained optimization. Adversarial debiasing or equality-of-opportunity constraints to ensure equitable predictions across demographic groups.

Bibliography

- [1] Breiman, L. (1996). Stacked Regressions. *Machine Learning*, 24(1), 49–64.
- [2] Chen, T., Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. *Proceedings of the 22nd ACM SIGKDD*, 785–794.
- [3] Cox, D.R. (1958). The Regression Analysis of Binary Sequences. *Journal of the Royal Statistical Society: Series B*, 20(2), 215–242.
- [4] Friedman, J.H. (2001). Greedy Function Approximation: A Gradient Boosting Machine. *Annals of Statistics*, 29(5), 1189–1232.
- [5] Hastie, T., Tibshirani, R., Friedman, J. (2009). *The Elements of Statistical Learning* (2nd ed.). Springer.
- [6] Ke, G., Meng, Q., Finley, T., et al. (2017). LightGBM: A Highly Efficient Gradient Boosting Decision Tree. *Advances in Neural Information Processing Systems*, 30, 3146–3154.
- [7] Lin, T.Y., Goyal, P., Girshick, R., He, K., Dollár, P. (2017). Focal Loss for Dense Object Detection. *IEEE International Conference on Computer Vision*, 2980–2988.
- [8] Lundberg, S.M., Lee, S.I. (2017). A Unified Approach to Interpreting Model Predictions. *Advances in Neural Information Processing Systems*, 30, 4765–4774.
- [9] Platt, J. (1999). Probabilistic Outputs for Support Vector Machines. *Advances in Large Margin Classifiers*, 61–74.

- [10] Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A.V., Gulin, A. (2018). CatBoost: Unbiased Boosting with Categorical Features. *Advances in Neural Information Processing Systems*, 31, 6638–6648.
- [11] Rubin, D.B. (1976). Inference and Missing Data. *Biometrika*, 63(3), 581–592.
- [12] Siddiqi, N. (2006). *Credit Risk Scorecards: Developing and Implementing Intelligent Credit Scoring*. John Wiley & Sons.
- [13] Viola, P., Jones, M. (2001). Rapid Object Detection Using a Boosted Cascade of Simple Features. *IEEE CVPR*, 1, 511–518.
- [14] Zadrozny, B., Elkan, C. (2002). Transforming Classifier Scores into Accurate Multiclass Probability Estimates. *Proceedings of the 8th ACM SIGKDD*, 694–699.